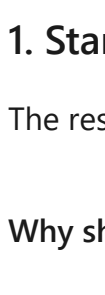




How does a Quant Researcher actually discover a New Trading Signal at a large hedge fund?



Mehul Mehta

Lead Quant at OCC, USA || Quant Finance (7+ Years) || 70K+ Followers||

Charles Schwab || PwC || Derivatives Pricing || Stochastic Calculus || Ris...



September 12, 2026

Many people imagine quantitative research as a simple process:

Download data → Train a machine-learning model → Generate a high Sharpe ratio → Deploy the strategy

In practice, this is often how overfitting begins.

At a serious systematic investment firm, discovering a new trading signal requires far more than finding an attractive historical pattern. Researchers must determine whether the signal is economically sensible, statistically robust, operationally feasible, scalable, and profitable after costs.

Here is what the end-to-end signal research process typically looks like.

1. Start With an Economic or Market Hypothesis

The research process usually begins with a question:

Why should this signal exist?

For example:

- Do stocks with improving earnings expectations outperform?
- Does momentum persist because investors react slowly to new information?
- Does deteriorating liquidity predict lower future returns?
- Can order-book imbalances reveal short-term buying or selling pressure?
- Do companies with improving profitability outperform similar companies?

A historical relationship without an economic, behavioral, or structural explanation may simply be a coincidence. The hypothesis gives the researcher a reason to believe that the signal could persist beyond the backtest.

2. Define the Prediction Target

Before building the model, the researcher must define exactly what the signal is expected to predict.

Possible targets include:

- Next-day returns
- Five-day forward returns
- One-month excess returns
- Probability of outperforming a benchmark
- Future volatility
- Downside risk
- Default probability
- Trading volume or liquidity

The prediction horizon must also match the intended trading strategy. A feature designed to predict returns over one month may not be useful for an intraday strategy.

Poorly defined targets often lead to misleading model results and incorrect portfolio decisions.

3. Build a Point-in-Time Dataset

Data preparation is often the most time-consuming part of quantitative research.

Researchers may combine:

- Prices and returns
- Company fundamentals
- Earnings announcements
- Analyst estimates and revisions
- Trading volume
- Liquidity measures
- Macroeconomic variables
- News and textual data
- Alternative datasets
- Order-book and market-microstructure data

The central question is not merely whether the data exists today.

The important question is:

Was this information actually available when the investment decision would have been made?

For example, using a company's revised financial statement before its publication date introduces look-ahead bias. A realistic dataset must reproduce what the researcher could genuinely have known at every historical point.

4. Clean and Validate the Data

Financial data is noisy, incomplete, and constantly changing.

Researchers must identify and address:

- Missing observations
- Extreme outliers
- Stock splits and corporate actions
- Stale prices
- Incorrect timestamps
- Delisted securities
- Survivorship bias
- Look-ahead bias
- Restated financial statements
- Changes in index membership
- Differences across data vendors

A sophisticated model cannot compensate for contaminated data. In many cases, a strong backtest is caused by a data issue rather than a genuine investment opportunity.

This is why institutional research teams spend significant time building data-quality checks before modeling begins.

5. Engineer Economically Meaningful Features

Raw data rarely becomes a usable signal without transformation.

Researchers convert raw variables into features that represent economically meaningful ideas:

- Prices → Momentum or mean-reversion features
- Financial statements → Value, quality, profitability, or leverage factors
- Analyst estimates → Earnings-revision signals
- Price and volume → Liquidity or investor-flow measures
- Order-book data → Imbalance and market-pressure indicators
- Options data → Implied volatility, skew, or tail-risk features
- Macroeconomic data → Growth, inflation, or monetary-policy regimes

Feature engineering is where financial intuition and statistical knowledge meet.

A feature should not merely improve model performance. Researchers must understand what it measures, why it should predict the target, and under what conditions the relationship could weaken.

6. Establish a Simple Baseline

Before using neural networks, transformers, or gradient-boosting models, researchers usually establish a simple benchmark.

Typical baseline methods include:

- Rank correlation
- Information Coefficient
- Linear regression
- Logistic regression
- Quantile portfolios
- Simple long-short portfolios
- Traditional factor models

For example, stocks may be ranked by an earnings-revision score. The researcher can then compare the returns of the highest-ranked stocks with those of the lowest-ranked stocks.

If a simple model cannot identify a stable relationship, a complex model may only be fitting noise.

Complexity should add measurable value not conceal a weak hypothesis.

7. Test Statistical Significance and Signal Stability

A profitable backtest does not automatically prove that a signal is real.

Researchers evaluate:

- Average Information Coefficient
- Information Coefficient stability
- IC Information Ratio
- t-statistics
- p-values
- Hit rate
- Return monotonicity across portfolios
- Signal decay
- Turnover
- Cross-sectional consistency
- Time-series stability

They may also divide securities into quintiles or deciles based on signal strength. A credible signal should ideally show a reasonably consistent relationship between signal ranking and subsequent performance.

Researchers must also account for multiple testing. If thousands of features are tested, some will appear successful purely by chance.

One impressive backtest is not enough. Repeated robustness across samples and specifications provides stronger evidence.

8. Conduct Proper Out-of-Sample Testing

Financial datasets should not normally be split randomly like conventional machine-learning datasets.

Random splitting can allow information from the future to influence the training process.

Instead, researchers use techniques such as:

- Walk-forward validation
- Rolling-window estimation
- Expanding-window estimation
- Purged cross-validation
- Embargo periods
- Strict chronological train-validation-test splits

For example, a model may be trained using data from 2010–2017, validated on 2018–2019, and tested on 2020. The window is then moved forward and the process is repeated.

This simulates how the model would have been developed and updated in real time.

The final test period should remain untouched until major research choices have been completed. Repeatedly checking and optimizing against the test set turns it into another training dataset.

9. Neutralize Unwanted Risk Exposures

A supposed new alpha signal may actually be a disguised exposure to an existing factor.

The signal may unintentionally favor:

- Small-cap stocks
- Value stocks
- High-momentum stocks
- Particular sectors
- Illiquid securities
- High-beta stocks
- Highly volatile companies

Researchers therefore test whether the signal still predicts returns after controlling for established factors.

A signal that disappears after sector and style neutralization may not contain independent information. It may simply repack an existing risk premium.

The objective is to measure the signal's **incremental information**, not just its raw historical return.

10. Evaluate Portfolio-Level Value

A signal can have a strong Information Coefficient and still be commercially useless.

Implementation matters.

Researchers evaluate:

- Expected return
- Expected volatility
- Sharpe ratio
- Maximum drawdown
- Turnover
- Transaction costs
- Bid-ask spread
- Slippage
- Market impact
- Borrowing costs
- Short availability
- Capacity
- Crowding
- Portfolio correlation
- Marginal contribution to risk

A fast-decaying signal may appear attractive before costs but become unprofitable after realistic execution assumptions.

Likewise, a signal may work for a \$1 million portfolio but fail when scaled to \$1 billion because the required trades move the market.

Therefore, the real question is not simply:

Does the signal predict returns?

It is:

Does the signal improve the portfolio after accounting for costs, constraints, capacity, and risk?

11. Stress-Test the Signal

Researchers test whether the signal survives changes in market conditions and research assumptions.

They may evaluate performance across:

- Bull and bear markets
- High- and low-volatility periods
- Inflationary and recessionary regimes
- Different countries
- Different sectors
- Large-, mid-, and small-cap stocks
- Liquid and illiquid securities
- Alternative formation periods
- Alternative holding periods
- Different portfolio-rebalancing frequencies
- Different transaction-cost assumptions

Researchers may also slightly change the parameters. If performance collapses when a 20-day window becomes a 21-day window, the result may be excessively optimized.

A robust signal should not depend on one unusually precise parameter combination or one favorable historical period.

12. Combine the Signal With Existing Alphas

Large systematic funds rarely deploy signals independently.

A new signal is evaluated alongside the firm's existing alpha library.

Researchers examine its:

- Correlation with existing signals
- Incremental portfolio return
- Incremental Sharpe ratio
- Diversification benefit
- Contribution to portfolio risk
- Effect on turnover
- Effect on transaction costs
- Capacity usage

A signal with a moderate standalone Sharpe ratio may be extremely valuable if it performs during periods when existing signals struggle.

Conversely, a strong standalone signal may add little value if it is highly correlated with signals already in production.

Institutional investors therefore care about the signal's **marginal contribution to the complete portfolio**.

13. Run a Paper-Trading or Shadow Portfolio

Before allocating real capital, the strategy may be operated as a live shadow portfolio.

It generates trades using real-time data but does not necessarily execute them with actual capital.

Researchers monitor:

- Whether live data arrives correctly
- Signal stability
- Trading frequency
- Data revisions and delays
- Expected versus realized transaction costs
- Slippage
- Execution feasibility
- Capacity
- Unexpected factor exposures
- Differences between research and production code

Paper trading helps identify problems that cannot be observed in a historical backtest.

A model may work perfectly with cleaned historical data but fail when exposed to delayed feeds, missing observations, changing security identifiers, or real execution constraints.

14. Move Toward Production Deployment

Only after passing research, validation, portfolio, risk, and implementation reviews can the signal move toward production.

The overall pipeline may look like this:

Economic hypothesis

↓

Point-in-time data construction

↓

Data cleaning and validation

↓

Feature engineering

↓

Signal construction

↓

Statistical testing

↓

Out-of-sample validation

↓

Risk and factor neutralization

↓

Transaction-cost and capacity modeling

↓

Portfolio construction

↓

Paper trading

↓

Production deployment

↓

Continuous monitoring

Deployment is not the end of the research process.

Once live, the signal must be continuously monitored for:

- Performance deterioration
 - Changes in IC
 - Rising turnover
 - Increasing transaction costs
 - Capacity constraints
 - Factor drift
 - Data-quality problems
 - Changes in market structure
 - Model instability
- If the economic relationship weakens or implementation costs increase, the firm may reduce the signal's weight, recalibrate it, or retire it entirely.
- Final Takeaway**
- The biggest misconception about quantitative research is that the job is to find a profitable historical pattern.
- It is not.
- The real job is to determine whether that pattern contains information that is:
- Economically explainable
 - Statistically reliable
 - Stable out of sample
 - Distinct from existing factors
 - Profitable after transaction costs
 - Scalable across meaningful capital
 - Compatible with portfolio constraints
 - Robust enough to survive changing markets
- That is the difference between an attractive backtest and a real institutional trading signal.
- A backtest shows what might have worked in the past.
- A rigorous quant research process determines whether the signal deserves to risk actual capital.